

120 Days of Quantitative Finance

Month 5: Advanced Quant Systems

Week 17

Day 118: Volatility Trading Strategies

1 Introduction

Volatility is not only a parameter in option pricing models but also a tradable asset.

Volatility trading strategies aim to exploit discrepancies between implied volatility and expected future volatility.

2 Implied vs Realized Volatility

- Implied volatility (σ_{imp}): extracted from option prices
- Realized volatility (σ_{real}): observed from historical returns

Key Idea:

$$\sigma_{imp} \neq \sigma_{real}$$

This difference creates trading opportunities.

3 Volatility Risk Premium

Empirically:

$$\sigma_{imp} > \sigma_{real}$$

This difference is known as the volatility risk premium.

4 Long Volatility Strategies

4.1 Straddle

Buy call and put at the same strike:

- profits from large price movements
- sensitive to volatility increase (positive Vega)

4.2 Strangle

Buy out-of-the-money call and put:

- cheaper than straddle
- requires larger moves

5 Short Volatility Strategies

5.1 Short Straddle

Sell call and put:

- profits from low volatility
- exposed to large losses during extreme moves

5.2 Iron Condor

Combine spreads to limit risk:

- collects premium
- profits if price remains stable

6 Role of Greeks

6.1 Delta

- measures directional exposure
- hedged to isolate volatility exposure

6.2 Gamma

- measures convexity
- long gamma benefits from large moves

6.3 Vega

- measures sensitivity to volatility
- primary driver in volatility trading

7 Delta Hedging

To isolate volatility:

- maintain $\Delta \approx 0$
- rebalance dynamically

8 P&L Decomposition

$$\text{P\&L} \approx \text{Vega} \cdot \Delta\sigma + \frac{1}{2}\Gamma \cdot (\Delta S)^2$$

9 Trading Signals

- compare implied vs forecasted volatility (GARCH)
- exploit mispricing

10 Risks

- jump risk
- liquidity risk
- model risk

11 Conclusion

Volatility trading transforms option pricing into actionable strategies, leveraging Greeks to manage and profit from market uncertainty.